Fig. 1: Overall System Flowchart

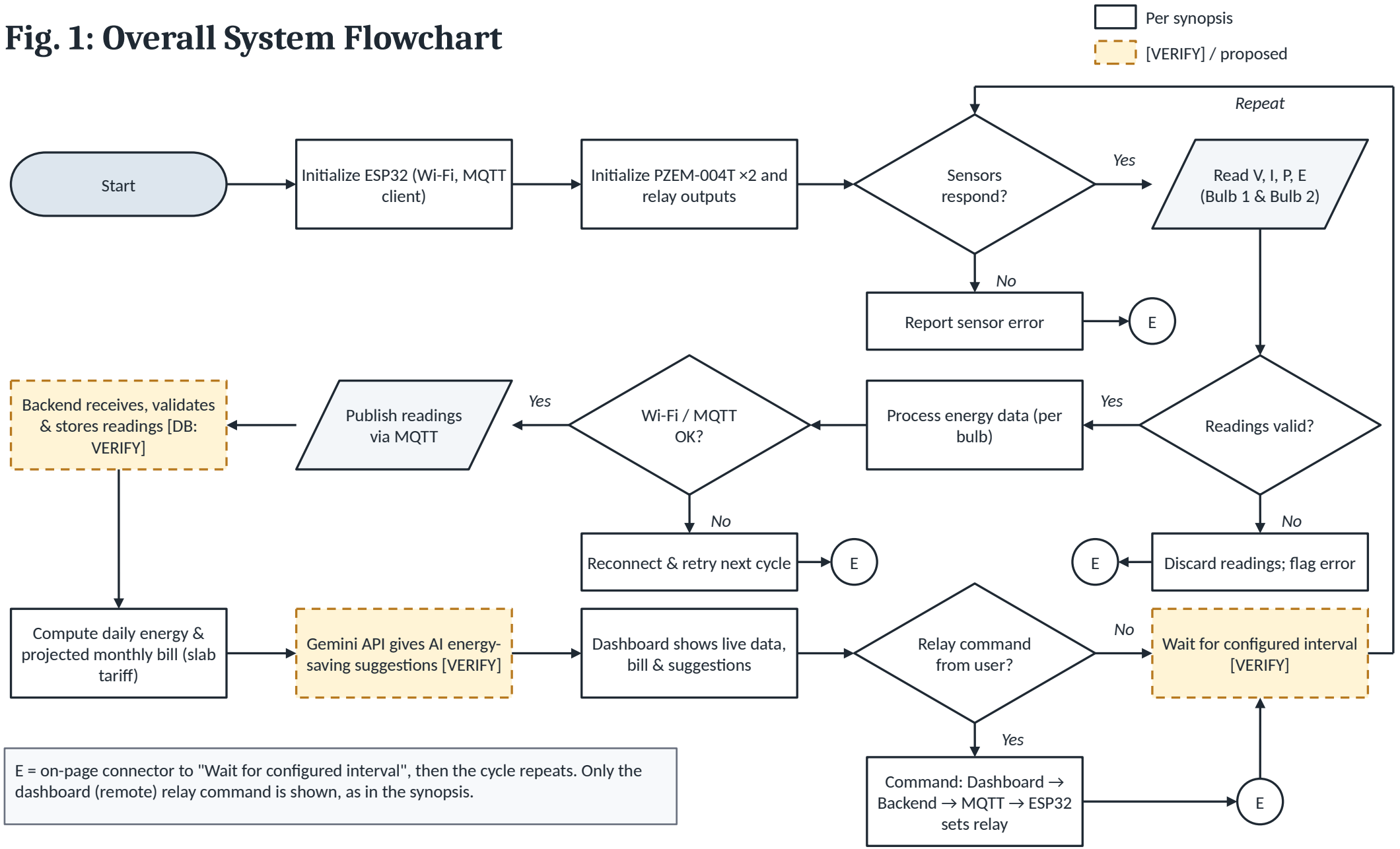


Fig. 1: Overall system flowchart of the Home Energy Optimizer, from PZEM-004T sensing on the ESP32 to MQTT transmission, backend processing, dashboard display, AI suggestions and remote relay control.

DRAFT: based on the project synopsis; confirm against the actual implementation. [VERIFY] = [VERIFY WITH ACTUAL IMPLEMENTATION]

Fig. 2: Sensor-Reading Flowchart

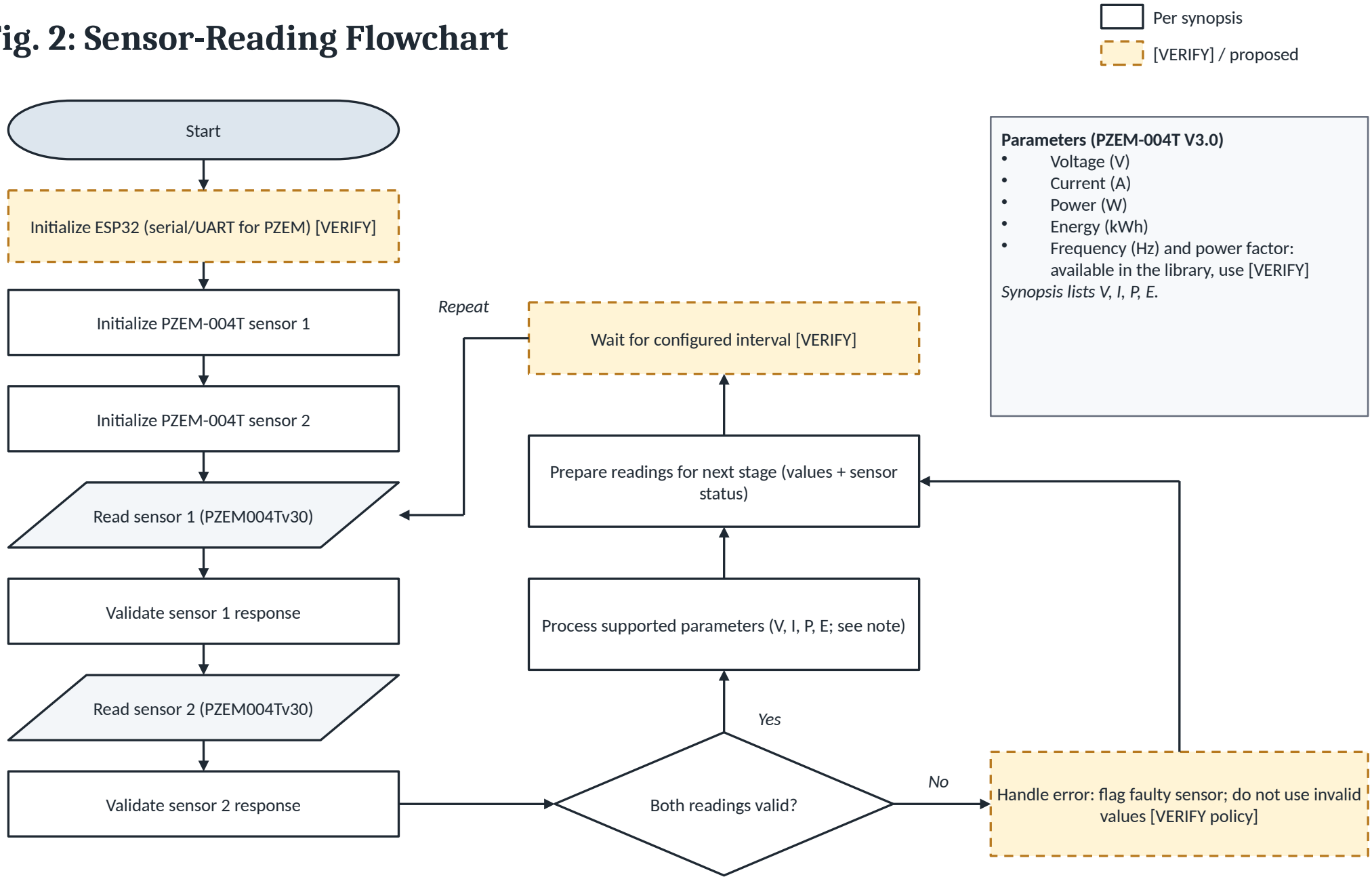


Fig. 2: Sensor-reading flowchart showing sequential reading and validation of the two PZEM-004T V3.0 sensors by the ESP32 and preparation of the readings for transmission.

DRAFT: based on the project synopsis; confirm against the actual implementation. [VERIFY] = [VERIFY WITH ACTUAL IMPLEMENTATION]

Fig. 3: Data Communication Flowchart

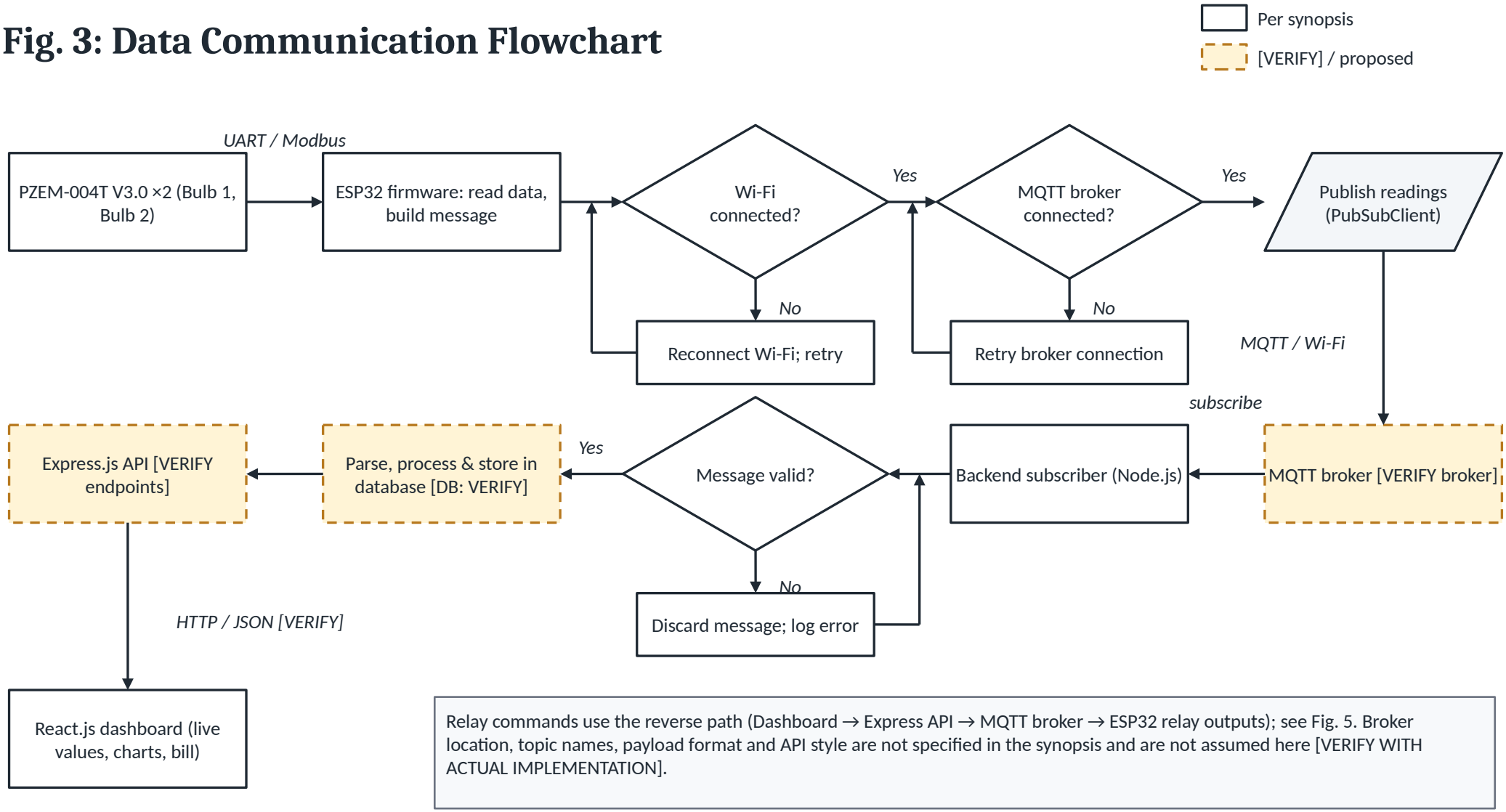


Fig. 3: Data communication flowchart showing the path of readings from the PZEM sensors through the ESP32, MQTT broker and backend to the React.js dashboard, with Wi-Fi, MQTT and message-validity error handling.

DRAFT: based on the project synopsis; confirm against the actual implementation. [VERIFY] = [VERIFY WITH ACTUAL IMPLEMENTATION]

Fig. 4: Electricity Bill Calculation Flowchart

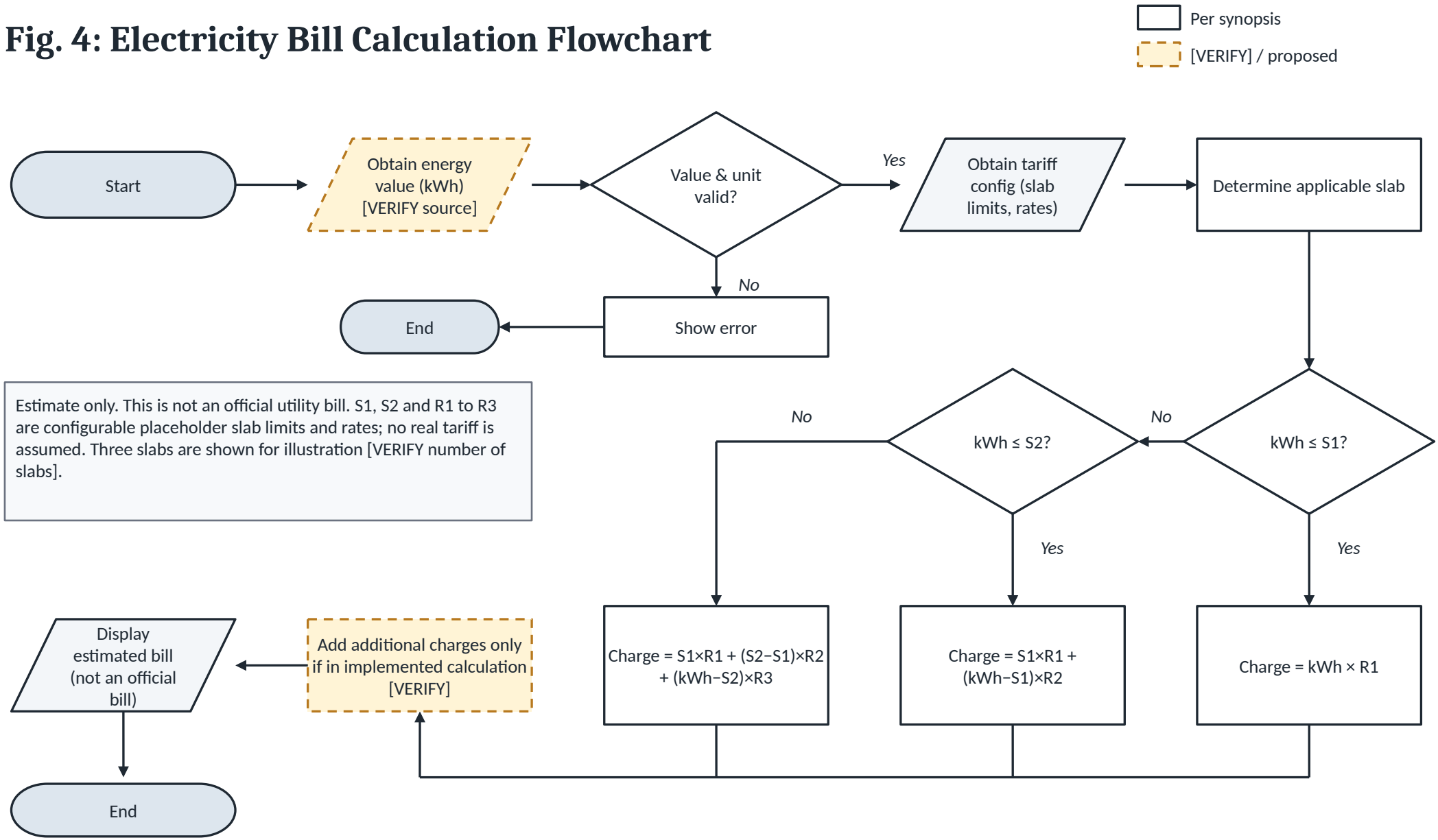


Fig. 4: Electricity bill estimation flowchart using configurable slab-wise tariff rates (placeholder values; the result is an estimate, not an official utility bill).

Fig. 5: Relay-Control Flowchart

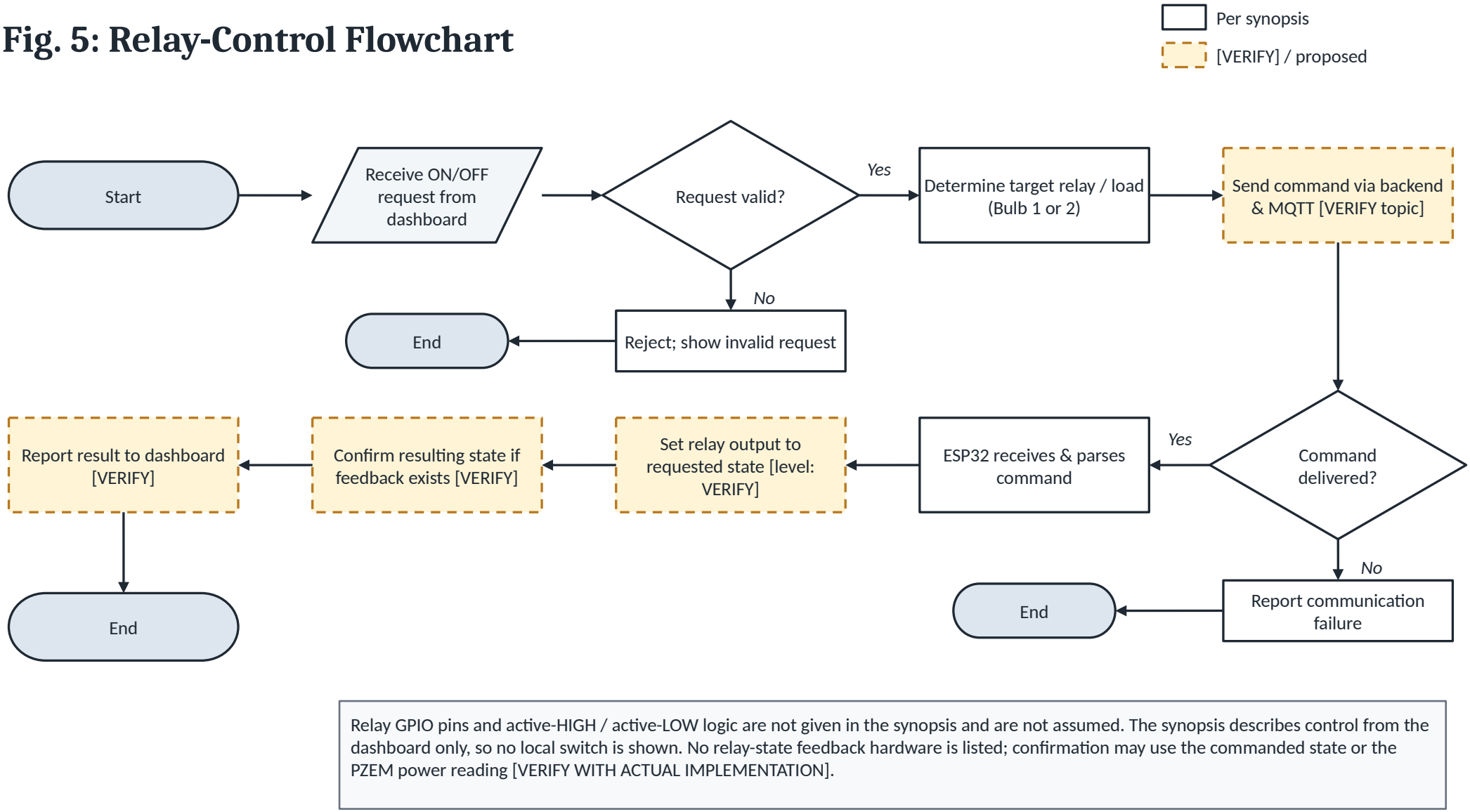


Fig. 5: Relay-control flowchart showing a dashboard ON/OFF request travelling through the backend and MQTT to the ESP32, which switches the selected relay channel.

Fig. 6: AI Energy-Saving Recommendation Flowchart

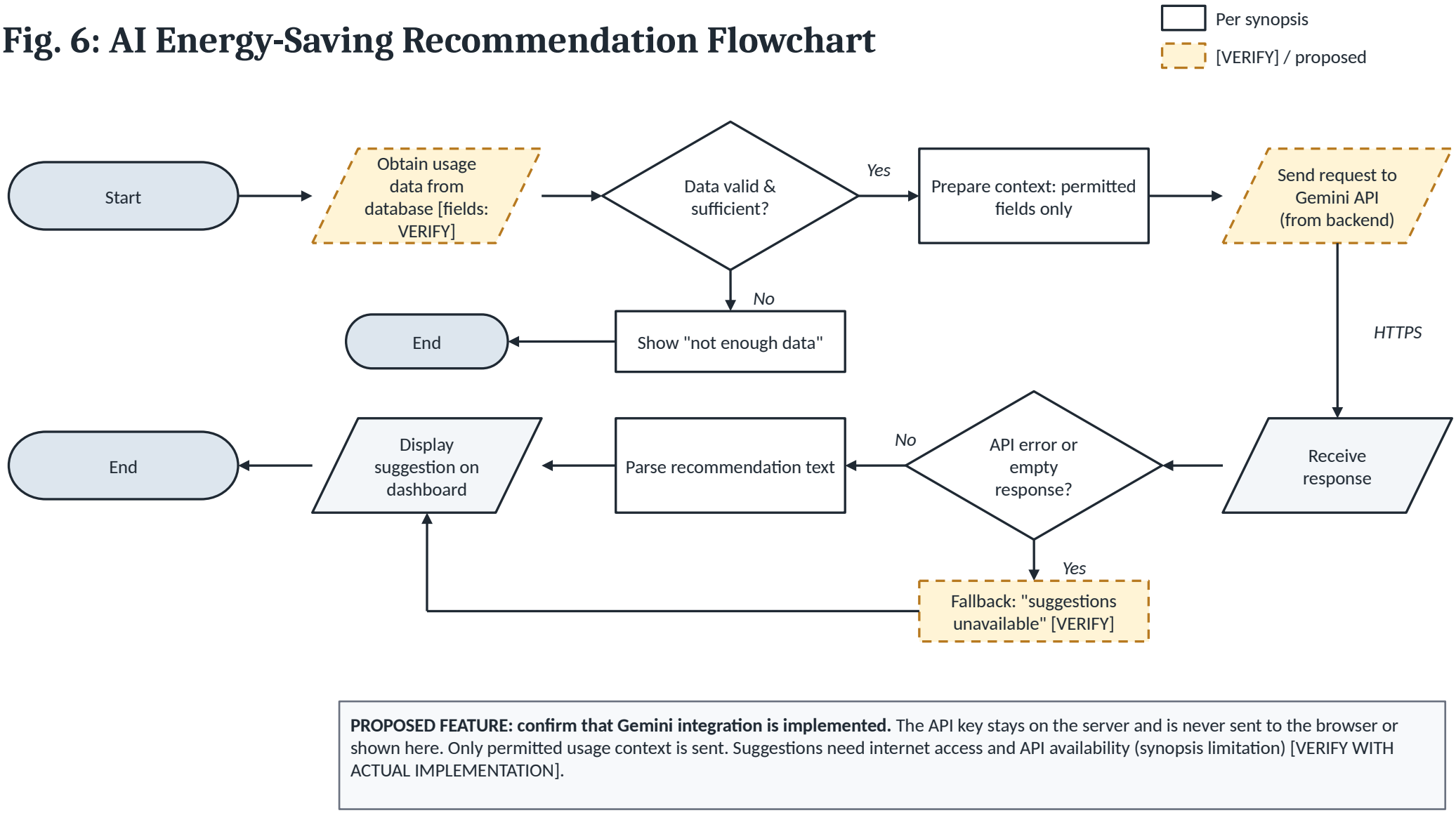


Fig. 6: AI energy-saving recommendation flowchart using the Gemini API, with a fallback when the API is unavailable (proposed feature: confirm implementation status).

Fig. 7: Use-Case Diagram

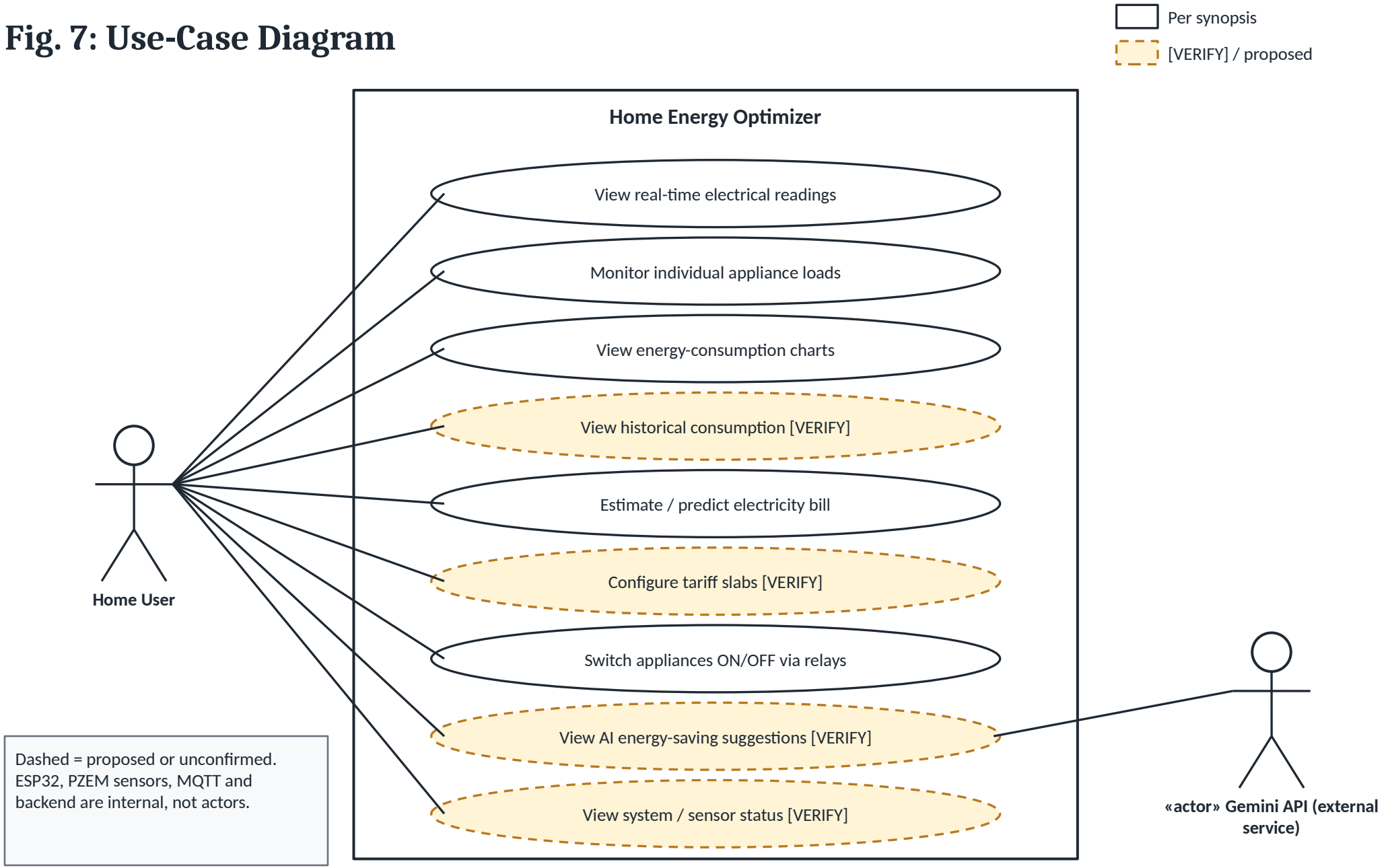


Fig. 7: UML use-case diagram of the Home Energy Optimizer showing the Home User and the Gemini API (external service) as actors.

Fig. 8: Project Gantt Chart (Proposed Schedule)

No	Phase	Start	Dur.	Done	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
1	Project selection and synopsis	1	1	0%												
2	Literature review	1	2	0%												
3	Requirement analysis	1	1	0%												
4	Hardware component procurement	2	2	0%												
5	Circuit assembly and sensor testing	4	2	0%												
6	ESP32 firmware development	5	3	0%												
7	Communication and backend development	7	2	0%												
8	Database integration	7	2	0%												
9	Dashboard development	2	2	0%												
10	Electricity bill calculation	8	1	0%												
11	Gemini API integration (if implemented)	9	1	0%												
12	System integration	9	2	0%												
13	Testing and troubleshooting	9	3	0%												
14	Documentation and blackbook preparation	11	2	0%												
15	Final review and demonstration	12	1	0%												

Follows synopsis weeks Placeholder weeks [VERIFY] **PROPOSED SCHEDULE: edit start week, duration and completion in the Excel file.**

Fig. 8: Proposed project schedule (Gantt chart) in weeks for the Home Energy Optimizer. Weeks are relative to the project start; no calendar dates are assumed.